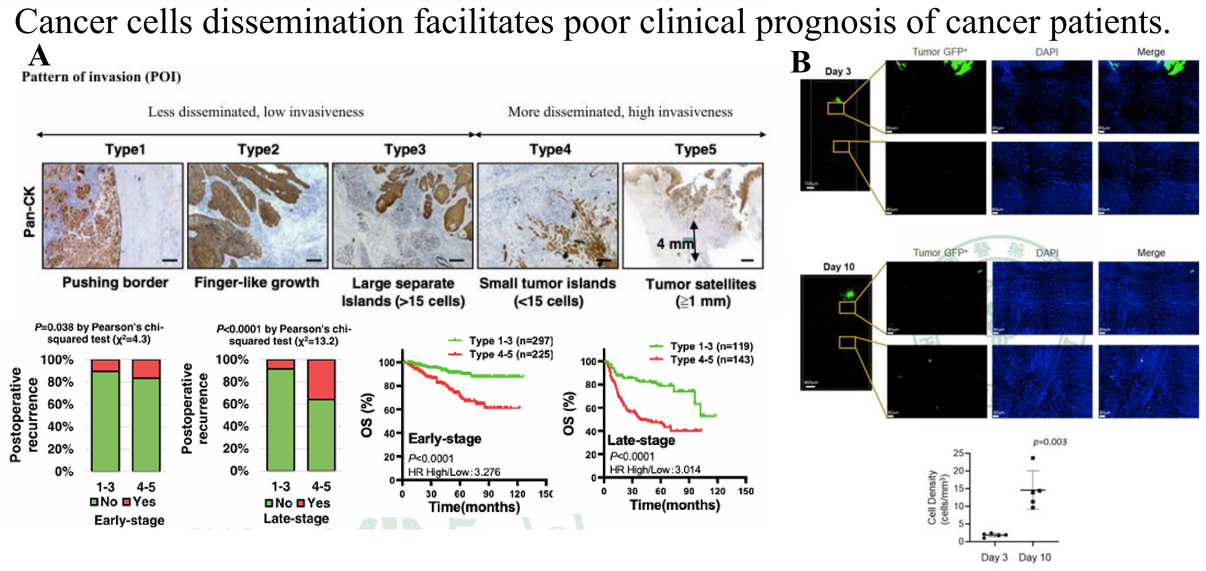
Development of Peptide Blockers for Preventing OSCC Recurrence through Disrupting Nerve-Tumor Communication

Chang-Hung Hsieh[1], Yen-Tang Chan[1], Yun-Ting Yeh[1], Yung-Chun Chang[1], Heng-Hsiung Wu[2]

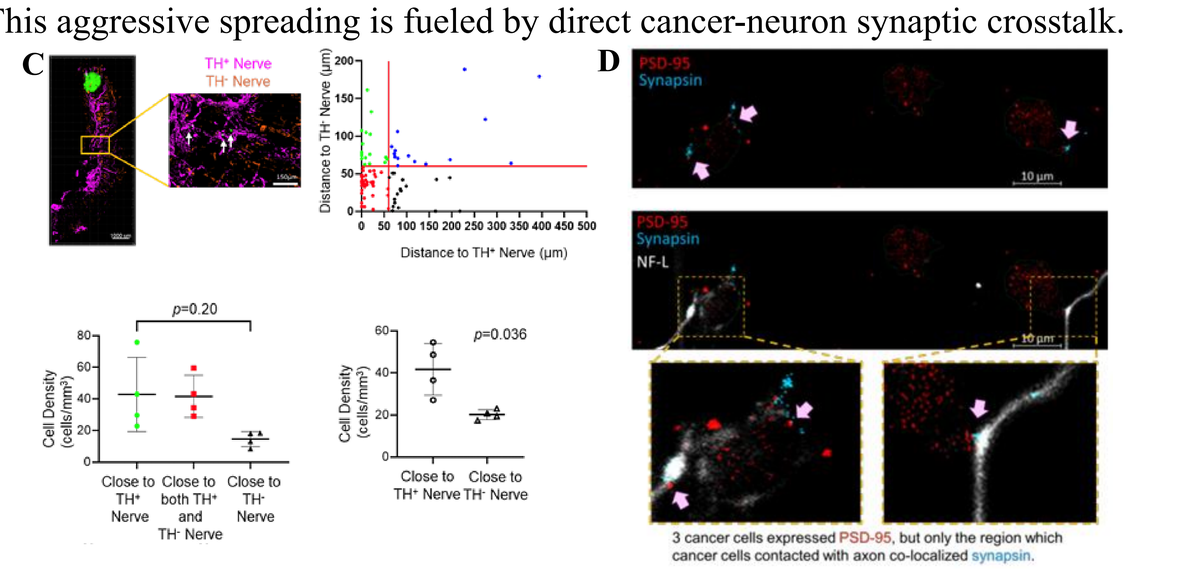
[1]School of Medicine, College of Medicine, China Medical University, Taichung, Taiwan
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Introduction

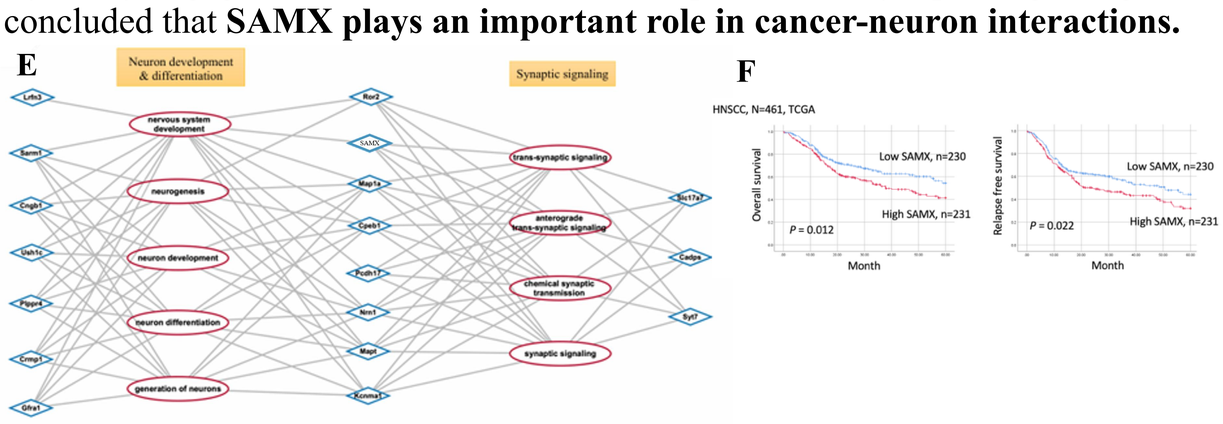
Cancer cells dissemination facilitates poor clinical prognosis of cancer patients.



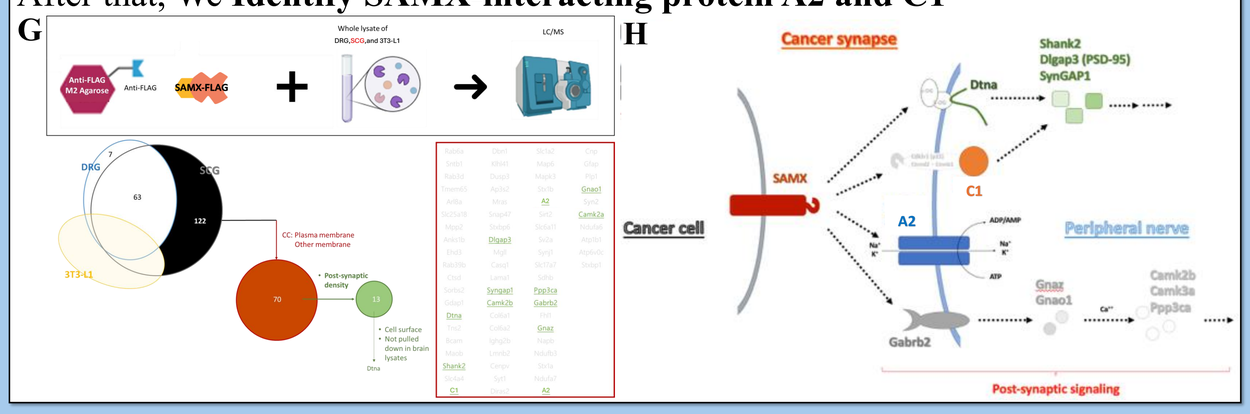
This aggressive spreading is fueled by direct cancer-neuron synaptic crosstalk.



By analyzing genes related to neuron development and synaptic signaling, we concluded that **SAMX plays an important role in cancer-neuron interactions.**



After that, We **Identify SAMX-interacting protein A2 and C1**



Aims

After targeting SAMX as the key mediator of cancer-communication, we aim to:

1. Identify the SAMX–ligand binding interface by Alphafold3
2. Design and prioritize SAMX-targeting peptide blockers by Bindcraft
3. Validate peptide blockers in a tumor–nerve co-culture model

References

- N. Vigneswaran and M. D. Williams, "Epidemiological Trends in Head and Neck Cancer and Aids in Diagnosis," Oral Maxillofac. Surg. Clin. N. Am., vol. 26, no. 2, pp. 123–141, May 2014, doi: 10.1016/j.coms.2014.01.001.
- Chen, Y.-T. (2023). *To investigate the roles of nerves in head and neck squamous cell carcinoma* [Master's thesis, China Medical University].
- Chen, K.-T. (2024). *The role of Neurexin 2 in the interactions between oral cancer cells and nerves* [Master's thesis, China Medical University].
- "Using AlphaFold Multimer to discover interkingdom protein–protein interactions - Homma - 2024 - The Plant Journal - Wiley Online Library." Accessed: Jan. 27, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/full/10.1111/tpj.16969

Methodology

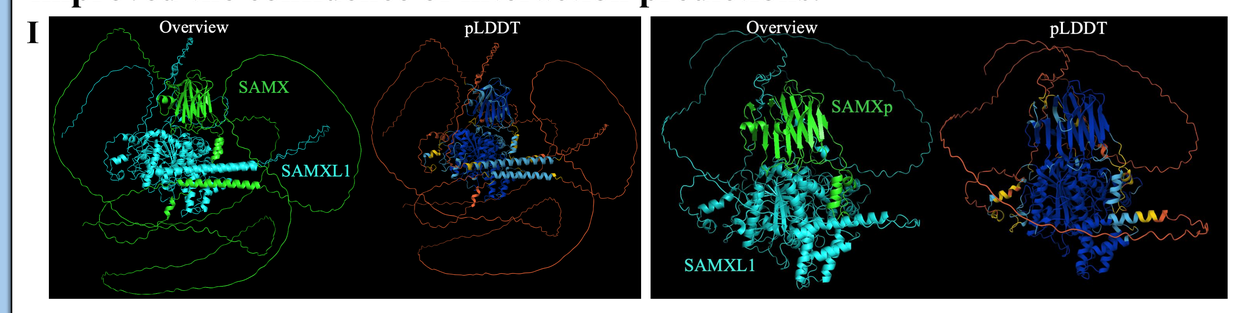
AlphaFold3 model SAMX interactions with defined ligands (SAMXL1/2) and candidate ligands (A2/C1) to identify shared binding interfaces

BindCraft design peptide blockers targeting the predicted SAMX interface prioritize structural confidence, hydrophilicity and stability

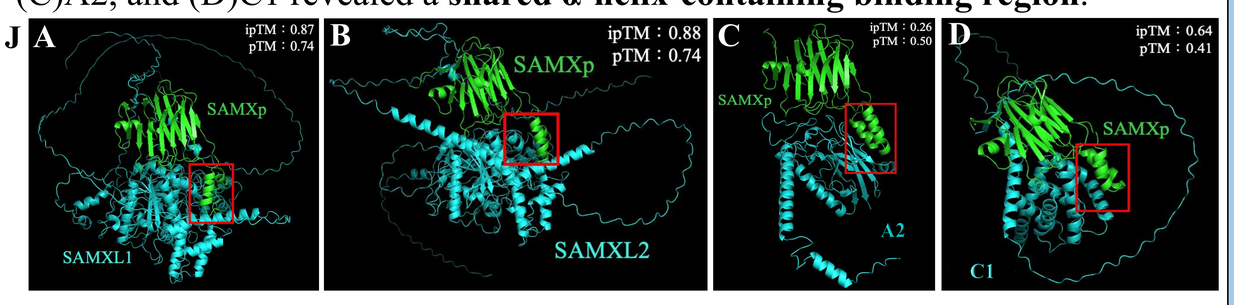
SCG-derived peripheral neuron–OSCC co-culture test

Preliminary Results

SAMX structure shows low confidence, so we remove the instability region to generate a more stable core structure termed **SAMXp, which substantially improved the confidence of interaction predictions.**



Subsequent modeling of SAMXp with its ligands (A)SAMXL1, (B)SAMXL2, (C)A2, and (D)C1 revealed a **shared α-helix-containing binding region**.

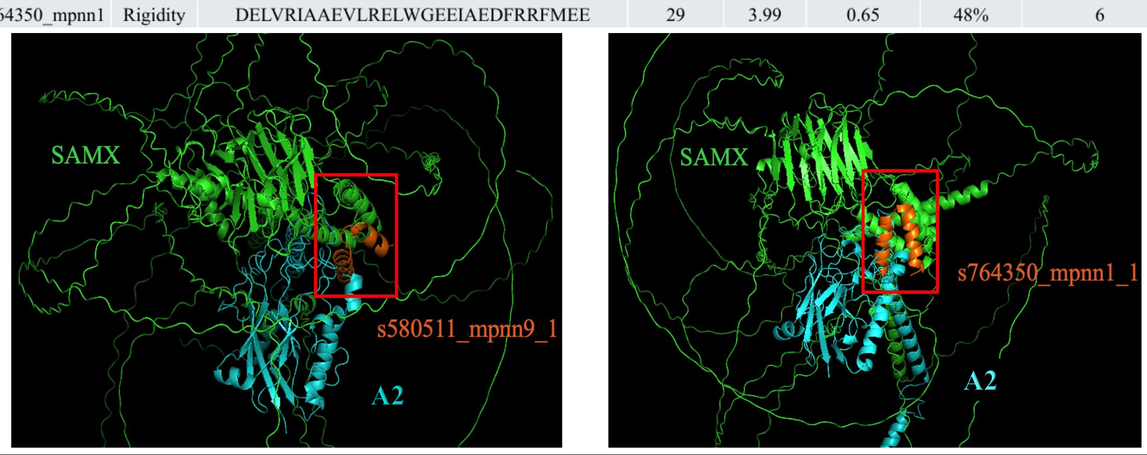


Targeting this α-helix region, we **develop two lead peptides based on structural confidence、hydrophilicity and stability**.

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Peptide	Mode	Sequence	Residues	Isoelectric point	Average hydrophilicity	Ratio of hydrophilic residues	Blocking C1 residues
s580511_mpnn9	Flexible	SEEEIKAKMDDIWQKNPANFFDDLIAWLS	29	3.87	0.26	52%	9
s764350_mpnn1	Rigidity	DELVRIAAEVLRELWGEEIAEDFRRFMEE	29	3.99	0.65	48%	6

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Future

